

THE LUNAR VERTEX PRISM PAYLOAD: UPDATES AND PLANS. David T. Blewett^{1,*}, Jasper Halekas², Peter Kollmann¹, Benjamin T. Greenhagen¹, Brett W. Denevi¹, Heather M. Meyer¹, Rachel L. Klima¹, Sarah K. Vines³, C. Dany Waller¹, Joshua T. Cahill¹, Brian J. Anderson¹, Leonardo Regoli¹, William F. Ames¹, Scott A. Cooper¹, Ann L. Cox¹, George C. Ho³, Jörg-Micha Jahn³, Lon L. Hood⁴, Sonia Tikoo⁵, Xiao-Duan Zou⁶, Mark Wieczorek⁷, Myriam Lemelin⁸, Shahab Fatemi⁹, Edward A. Cloutis¹⁰. ¹Johns Hopkins University Applied Physics Laboratory, Laurel, MD 20723, USA. ²Univ. of Iowa, Iowa City, IA. ³Southwest Research Inst., San Antonio, TX. ⁴Univ. of Arizona, Tucson, AZ. ⁵Stanford Univ., Stanford, CA. ⁶Planetary Science Inst., Tucson, AZ. ⁷Institut de Physique du Globe de Paris, France. ⁸Univ. de Sherbrooke, QC, Canada. ⁹Umeå Univ., Sweden. ¹⁰Univ. of Winnipeg, Winnipeg, MB, Canada. (*david.blewett@jhuapl.edu).

1. Introduction: *Lunar Vertex*, NASA's first Payloads and Research Investigations on the Surface of the Moon (PRISM) delivery, will explore a portion of Reiner Gamma (7.585° N, 58.725° W). Reiner Gamma (RG) is home to a magnetic anomaly, a region of magnetized crustal rocks. The RG magnetic anomaly is co-located with the type example of a class of irregular high-reflectance markings known as lunar swirls.

The *Lunar Vertex* payload will be carried to the lunar surface on a commercial lander as part of NASA's Commercial Lunar Payload Services (CLPS) program. The lander and payload are designed for operation during one lunar daylight period (i.e., no night operations or survival). The lander will arrive at the surface shortly after local sunrise, and the mission will end ~10–13 Earth days later. NASA selected Intuitive Machines (IM) as the CLPS provider for the Reiner Gamma delivery (known as the CLPS "CP-11" mission, which will be IM's third lunar lander). Launch will be no earlier than December 2025. Landing will be ~5-7 days after launch.

2. Science Background: RG is a natural laboratory for addressing a wide range of questions that touch on planetary magnetism, lunar geology, space plasma physics, and space weathering. Several hypotheses for the origin of lunar magnetic anomalies have been proposed (summarized in [1]), including remanent magnetism from an ancient lunar dynamo and magnetism caused by plasma effects related to ejecta from basin-forming impacts or by the impact of a comet.

Co-located with the RG magnetic anomaly is the most famous example of a lunar swirl. The origin of swirls is a long-standing puzzle. Hypotheses for the formation of swirls (summarized in [1]) include magnetic shielding of the surface from the darkening effects of the solar wind, scouring by gas and dust during the collision of a comet's coma or a meteoroid swarm, and accumulation of fine-grained dust in response to magnetic or electric fields associated with the magnetic anomaly.

The presence of the magnetic anomaly at RG causes interactions with the solar wind that lead to interesting plasma physics phenomena. For example, data from

orbital spacecraft have demonstrated that mini-magnetosphere stand-off regions exist above the Moon's strongest magnetic anomalies (cf. [1]).

The *Lunar Vertex* payload was designed to address three science goals [2, 3] (see Section 5 below). The science goals and objectives will be accomplished with instruments on the lander and on a small rover.

3. Lander Instruments: The three *Lunar Vertex* lander instruments were delivered to Intuitive Machines in June 2023. *Magnetic Anomaly Plasma Spectrometer (MAPS)*. MAPS, built by the Southwest Research Institute, will measure the ion and electron velocity distribution over a $292.5^\circ \times 90^\circ$ FOV from 8 eV to 17.5 keV energy per charge [4, 5].

Vector Magnetometer-Lander (VML). VML was built by APL. VML is comprised of a tetrahedral array of four commercial fluxgate magnetometers mounted at the bottom of a 0.5-meter mast, with a science-grade dual ring-core fluxgate magnetometer positioned at the top of the mast [4, 6, 7].

Vertex Camera Array (VCA). VCA, built by Redwire, is a set of three clusters of three RGB cameras [4, 8]. VCA images will be used to characterize the landing site geology and to understand the physical properties of the lunar regolith around the lander.

4. Rover and Rover Instruments: The rover vehicle (chassis) built by Lunar Outpost arrived at APL in the spring of 2023. The vehicle [4] is a variant of Lunar Outpost's Mobile Autonomous Prospecting Platform (MAPP) product. The rover instruments were integrated with the vehicle, and environmental testing was carried out on the integrated system. The integrated rover was delivered to Intuitive Machines in January 2024. Lunar Outpost will operate the rover on the Moon.

Vector Magnetometer-Rover (VMR). VMR [4, 6, 7], built by APL, is comprised of a tetrahedral array of four commercial fluxgate sensors mounted on a 20-cm mast on top of the rover.

Rover Multispectral Microscope (RMM). RMM [4, 9] is a close-up imager from Canadensys Aerospace, mounted inside the rover. It will provide information on soil texture. RMM can also measure spectral reflectance

using LED illumination at five UV to NIR wavelengths for determining regolith composition and maturity.

5. Science Goals and Traceability: The science goals and traceability to instrument measurements are summarized here. *Question 1:* What is the source of the Reiner Gamma magnetic anomaly? Is the anomaly the result of internal processes (e.g., magnetized by an ancient dynamo) or external processes related to basin-forming or comet impacts?

Objective 1.1: Determine vector B-field and intensity in a lunar magnetic anomaly – VML.

Objective 1.2: Measure B-field variability at various distances from lander – VMR.

Objective 1.3 Measure electron pitch-angle distribution to assess B-field topology – MAPS electron spectrometer.

Question 2: What process is responsible for the high-reflectance patterns called lunar swirls? Is magnetic shielding of the surface from space weathering by the solar wind the only cause of the bright swirls? Has the surface within the swirls undergone comet-related modification? Has unusual dust movement taken place?

Objective 2.1: Characterize site geology and regolith photometric properties proximal to and distant from the lander – VCA.

Objective 2.2: Characterize regolith micro-scale textural properties and composition within and outside of lander disturbance, in a profile extending toward the bright swirl/dark lane boundary – RMM.

Question 3: What is the structure of mini-magnetospheres that form over magnetic anomalies and how do they interact with the solar wind? What is the relationship between the plasma and magnetic field characteristics?

Objective 3.1: Characterize incident ions that reach lunar surface in an area with strong crustal B-field – MAPS ion spectrometer.

Objective 3.2: Characterize incident electrons that reach lunar surface in an area with strong crustal B-field – MAPS electron spectrometer.

6. Commissioning and Operations: *MAPS.* MAPS will conduct a power-on test and low-voltage checkout during cruise, and again after landing. After rover deployment, high-voltage (HV) commissioning with HV ramp limited to max energy 5 keV will be performed, followed by threshold data collection in this mode. Later, baseline HV commissioning (ramp to 17.5 keV max energy) will take place, and the instrument will operate in baseline HV mode for the rest of the mission.

VML. VML will power on and begin continuous data collection (10 vectors/s, vps) at a distance of ~10 Earth radii. It will continue in this mode through descent and landing. After rover deployment, the cadence will change to 1 vps through the end of mission.

VCA. Commissioning on the Moon will consist of boot-up and collection of one jpeg image per camera. For science data collection, jpegs will be downlinked to confirm exposure time. One set of raw science data images will be collected every 7 hr. throughout the mission. Additional image sets will be obtained to monitor rover activities or to observe special illumination conditions such as small phase angles.

Rover. Upon deployment, the rover will move to a site that is fully sunlit and within the field of view of VCA. The rover mission-success criteria are (a) One science stop in the immediate vicinity of the lander, and (b) One science stop at a location outside the area of lander disturbance (~20 m away, based on published relations [10, 11] and the dry mass of the IM Nova-C lander). Upon completion of these two initial stops, the rover will traverse generally north toward the transition between the bright portion of the swirl and the adjacent dark lane [3]. The rover must spend 2-3 hr. charging for each hour of operation. High temperatures may prevent rover operations past lunar noon.

VMR. A VMR aliveness check will be done during cruise. A functional check will take place after rover deployment. VMR will collect data during the rover's traverse from science stop 1 to stop 2, and from stop 2 to stop 3. At these and each subsequent science stop, VMR will collect 15 min. of data at 10 vps, with the rover "quiet".

RMM. An RMM aliveness check will be done during cruise. A functional check will take place after rover deployment. At the first two science stops, RMM will downlink one image for analysis of regolith texture. At later stops, RMM may collect both texture images and spectral image sets.

7. Lessons Learned: The PRISM program of rapid, low-cost, science investigations is managed under NASA NPR7120.8, which accepts higher risk and allows for substantially reduced oversight. *Lunar Vertex* is APL's first science mission developed under 7120.8 requirements. Lessons learned were reported by [12].

References: [1] D.T. Blewett et al. (2021), *B.A.A.S.* 53(4), DOI: 10.3847/25c2feb.9295af86. [2] D.T. Blewett et al. (2022), *LPSC 53rd*, abstr. 1131. [3] D.T. Blewett et al. (2023), *LPSC 54th*, abstr. 1780. [4] D.T. Blewett et al. (2024), *LPSC 55th*, abstr. 1553. [5] J.-M. Jahn et al. (2023), *LPSC 54th*, abstr. 2675. [6] C.D. Waller et al. (2023), *LPSC 54th*, abstr. 1820. [7] S.K. Vines et al. (2023), *LPSC 54th*, abstr. 2150. [8] B.W. Denevi et al., (2023), *LPSC 54th*, abstr. 2355. [9] R.L. Klima et al. (2023), *LPSC 54th*, abstr. 2718. [10] R.D. Lorenz (2016), *PSS* 127, 91. [11] R.N. Clegg-Watkins et al. (2016), *Icarus* 273, 84. [12] A.L. Cox & W.F. Ames (2024), *Small-Sat Conf.*, SSC24-III-10.